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🌐 Recovery of a Chromatographic Resin Used in the Prefractionation of Cyanobacterial Extracts

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We use this protocol and it's working

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Keywords: chromatography, fractionation, natural products, resin, prefractionation of cyanobacterial extracts hp20ss resin, cyanobacterial extracts hp20ss resin, fractionation of cyanobacterial extract, cyanobacterial extract, soxhlet extractor, using soxhlet extractor, chromatographic resin, organic solvent, divinylbenzene, recovery method for hp20ss, adsorption, natural product, sequence of organic solvent, porous adsorbent material, high effectiveness in the fractionation, efficient separation

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Disclaimer

The **Google Gemini** artificial intelligence tool was used to convert real laboratory photographs into stylized technical illustrations, allowing for the addition of clear annotations and the enhancement of component visualization.

Abstract

- HP20SS resin is a porous adsorbent material obtained through crosslinking between polystyrene and divinylbenzene. Its characteristics allow for the efficient separation of a wide range of compounds, in addition to the removal of impurities, through adsorption and size separation mechanisms. It stands out for its high effectiveness in the fractionation of natural products.
- This protocol aims to present a recovery method for HP20SS used in fractionation of cyanobacterial extracts, using Soxhlet extractor and a sequence of organic solvents.

Materials

Solvents and solutions:

- Extran
- Methanol (100%), ACS Grade;
- Ultrapure water, HPLC Grade;
- Acetone, ACS Grade;
- Ethyl Acetate, ACS Grade;
- Isopropanol (100%), HPLC Grade

Resin:

- DIAION™ HP20SS (used).

Equipment:

- SPE Column (used);
- Heating mantle;
- Soxhlet extraction cartridges;
- Soxhlet Extrator (with condenser)
- Round-bottom flask;
- Long spatula and tweezers.

Abstract

- 1
 - HP20SS resin is a porous adsorbent material obtained through crosslinking between polystyrene and divinylbenzene. Its characteristics allow for the efficient separation of a wide range of compounds, in addition to the removal of impurities, through adsorption and size separation mechanisms. It stands out for its high effectiveness in the fractionation of natural products.
 - This protocol aims to present a recovery method for HP20SS used in fractionation of cyanobacterial extracts, using Soxhlet extractor and a sequence of organic solvents.

Materials and Methods

- 2 Solvents and solutions:
 - Extran
 - Methanol (100%), ACS Grade;
 - Ultrapure water, HPLC Grade;
 - Acetone, ACS Grade;
 - Ethyl Acetate, ACS Grade;
 - Isopropanol (100%), HPLC Grade.

Resin:

- Used DIAION™ HP20SS (Mitsubishi Chemical Group Corporation).

Equipment:

- SPE Column (used);
- Heating mantle;
- Soxhlet extraction cartridges;
- Soxhlet extrator coupled to condenser
- Round-bottom flask;
- Long spatula and tweezers.

Removal of used resin and cleaning of columns

- 3 Using a spatula and long tweezers, remove the frits from the SPE column used after prefractionating cyanobacterial extracts (Jesus et al, 2025). This procedure is described below:
- 3.1
 - **Gently** tap the chromatograpgy column on a surface to dislodge the frits, taking care not to blow the packed solids to the surface. The step is described in Figure 1.

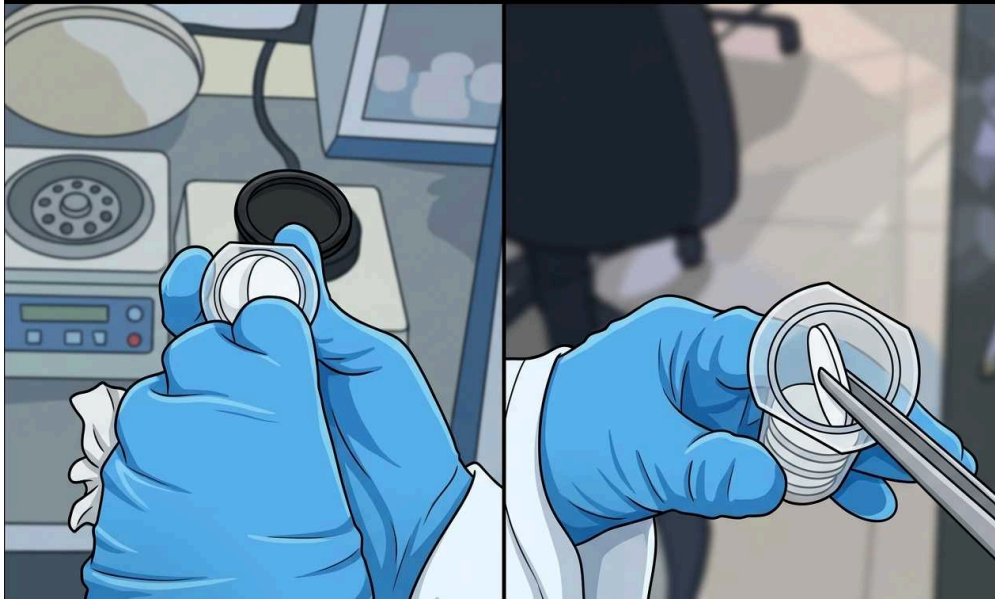
Figure 1: Demonstration of frit removal from a SPE column.



Source: Authors (2023).

- 3.2
- Use a spatula to tilt the frits and remove them with the long tweezers. This step is demonstrated in Figure 2:

Figure 2: Removing frits with spatula and tweezers.



Source: Authors (2023).

- If possible, do not let the extract load mix with the resin, so that recovery occurs more efficiently.
- 3.3
- Store the used HP20SS resin in a labeled bottle for recovery. It is recommended to accumulate resin used across multiple fractionation cycles prior to recovery.
- 3.4
- Once all frits have been removed from the columns, soak them in 100% isopropanol for approximately 1 hour. Allow the frits to dry and, subsequently, repeat the same procedure using acetone. Allow the frits to dry completely before reuse.
- 3.5
- Wash the chromatographic columns with Extran and rinse them with running water and Milli-Q water. Then rinse the columns with a little methanol to remove any possible Extran residues for later use in fractionation.

Chromatographic resin recovery

- 4 **This procedure must be carried out inside the fume hood due to the use of highly volatile organic solvents.** The procedure is described below:
- 4.1
- Wash the Soxhlet extraction cartridge with ethyl acetate and acetone before starting the procedure. To do this, place the cartridge in a 200 mL beaker and add the appropriate solvent so that it is completely covered. The cartridge should be soaked in each solvent for approximately 1 hour. The extraction cartridge is shown in Figure 3:

Figure 3: Soxhlet extraction cartridge.

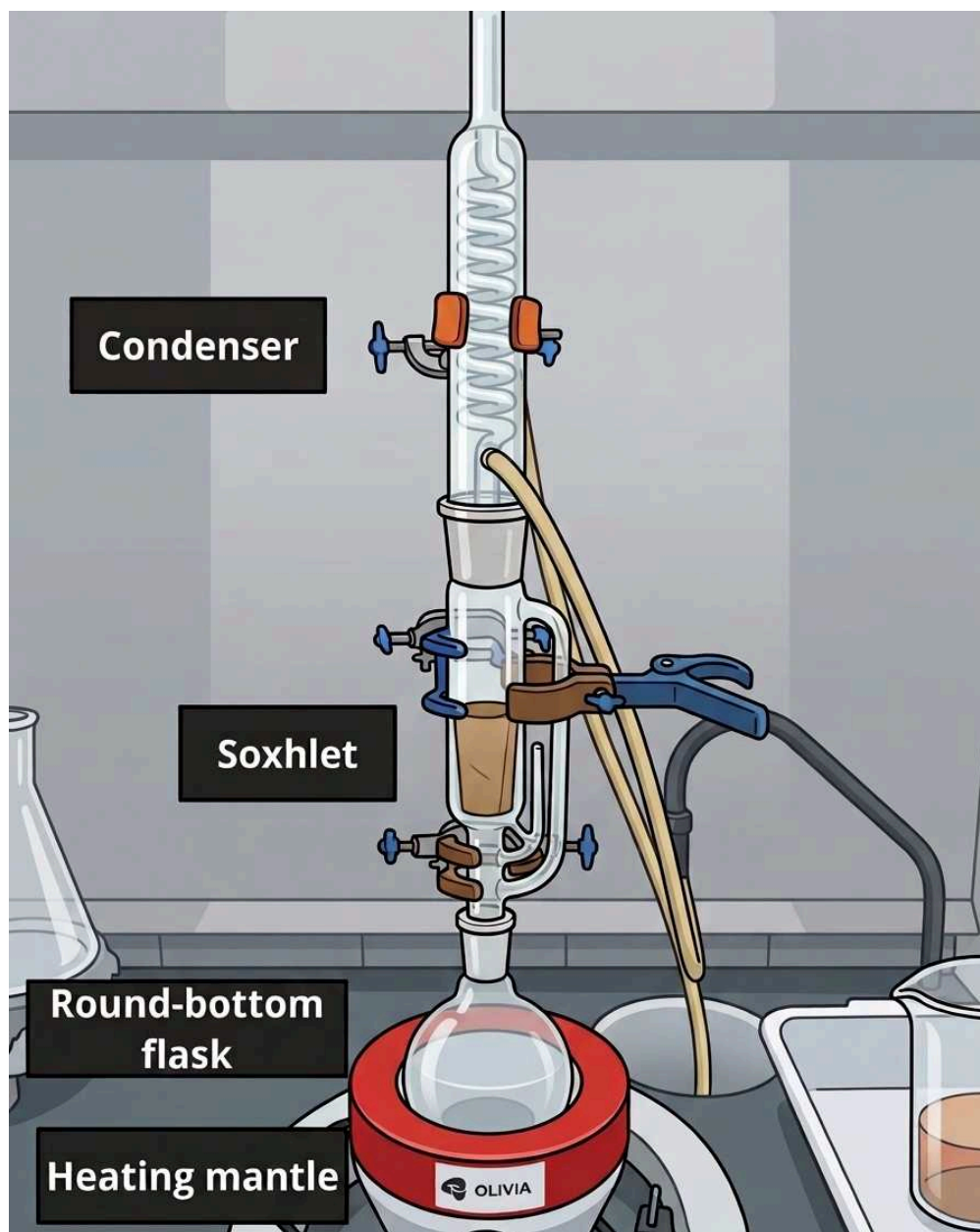


Source: Authors (2023).

- 4.2
- Add the used HP20SS resin to the cartridge to approximately 60% of its capacity and moisten the cartridge containing the resin with 100% isopropanol before introducing it into the Soxhlet. This is important because resin particles expand when in contact with the organic solvent and may overflow from the cartridge.

- 4.3 ■ Insert the cartridge with the resin moistened with 100% isopropanol into the Soxhlet extractor using long tweezers.
- 4.4 ■ Add 200 mL of 100% isopropanol to a 500 mL flat-bottomed or round-bottomed flask.
- 4.5 ■ Assemble the Soxhlet extractor, freshly cleaned with Extran soap and 100% methanol. The organization of the equipment is shown in figure 4 below:

Figure 4: Demonstration of the assembly of the Soxhlet extractor.



Source: Authors (2023).

- 4.6
- Fit rubber tubes for the running water inlet and outlet (the lower tube should be connected to the faucet and the upper tube to the sink for the running water outlet) and open the running water valve.
 - **The valve must be open and water must be circulating through the condenser throughout the procedure**
- 4.7
- Plug the heating mantle into the socket and set it to medium power. When bubbles appear, reduce the power a little and observe whether the organic solvent is slowly

evaporating.

- **Important: different heating mantles may behave differently.**

4.8 ▪ Extraction with 100% isopropanol should be performed within a range of **10 to 15 Soxhlet cycles.**

4.9 ▪ Once the extraction is complete, turn off the heating mantle and wait for the glassware to cool down and for the solvent evaporation to stop.

4.10 ▪ Replace the 100% isopropanol with 200 mL of acetone and repeat the extraction procedure.

- **Remember: it is necessary to moisten the resin with acetone in the cartridge before reintroducing it into the Soxhlet, just like the first extraction process with 100% isopropanol.**

4.11 ▪ Extraction with acetone should be carried **up to 10 Soxhlet cycles.**

4.12 ▪ After the extraction with acetone, turn off the heating mantle and wait until the glassware cools down.

4.13 ▪ Store used organic solvents and recovered resin in properly labeled flasks.

- **Organic solvents can be reused, as recovery via Soxhlet extraction involves continuous evaporation and condensation, effectively purifying the solvent during each cycle. Consequently, the presence of residual extract components in previously used ("contaminated") solvents does not compromise their suitability for subsequent extractions.**

- **Prior to storage, the resin should be spread in a tray and placed under a fume hood to allow complete evaporation of residual organic solvents from the extraction process.**

References

- 5 Jesus, L.S., Weiss, M.B., Yoon, J., do Amaral, S.C., Crnkovic, C.M. **Prefractionating Cyanobacterial Extracts by Solid-Phase.** *Protocols.io* (2025) dx.doi.org/10.17504/protocols.io.3byl4wdkrvo5/v1

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